# Collision lab

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Test momentum conservation with measured before-and-after velocities.

**SUCCESS CRITERIA - I can...**

1. Record masses, signs and velocities consistently.
2. Compare total momentum before and after.
3. Discuss external impulse and measurement variation.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

The same object stops from the same speed, but the stopping time is 3 times the original. Compare average stopping-force magnitudes.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $p_{total} = m_1v_1 + m_2v_2$; $\Delta p_{total} = p_{total, f} - p_{total, i}$. Compare just before/after collision. Use approved carts; record actual trials separately from practice data.

**Worked reasoning:** Supplied trials give total momenta 6.0 before and 5.8 kg·m/s after. The change is -0.2 kg·m/s. Compare this with measurement uncertainty before attributing it to a new physical effect.

**Your turn:** Draw a before-and-after momentum comparison and name one possible external interaction.

| Before | After |
|---|---|
| | |

Label mass, velocity and momentum.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Supplied case: cart A is 0.50 kg at +0.40 m/s; B is 0.50 kg at rest. After sticking, both move at +0.20 m/s. Find total momentum before and after.

| Cart | Before (m/s) | After (m/s) |
|---|---|---|
| A: 0.50 kg | +0.40 | +0.20 |
| B: 0.50 kg | 0 | +0.20 |

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Two 1 kg lab carts stick together. Before contact their velocities are +2 and 0 m/s; afterward both move at +1 m/s. Complete the table. Compare total momentum before and after and write the claim the data support.

| Cart | Mass (kg) | v before (m/s) | v after (m/s) |
|---|---|---|---|
| A | 1 | 2 | 1 |
| B | 1 | 0 | 1 |

**Marzano check - circle:** ① ② ③

## 05 Explain your reasoning

Design a table for three trials with masses and signed velocities before/after. Name one possible external impulse and one measurement limitation.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A launch has $v_x$=6 m/s and $v_y$, i=40 m/s upward. Use g=10 m/s². Find time to the top and total time back to launch height.
Label separate horizontal and vertical motion.



Add velocity / acceleration arrows.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Why must velocities be signed when carts approach each other?

| Object / trial | Mass | Velocity | Momentum |
|---|---|---|---|
| A | ____ | ____ | ____ |
| B | ____ | ____ | ____ |

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Group A has momentum change 0.1±0.2; group B has 0.8±0.2 kg·m/s. Which result is consistent with zero under the stated bounds?

| Object / trial | Mass | Velocity | Momentum |
|---|---|---|---|
| A | ____ | ____ | ____ |
| B | ____ | ____ | ____ |

**Marzano check - circle:** 1 2 3